MPG No. 24.3.89 6′22′ (6’22’ setting) [Type III, hexagonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & 0 \\ -T_{xxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -T_{xxx} & 0 \end{bmatrix}$$

$T_{xxx} = \mathbb{T}_{3c}^{(1)}$

* Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & -T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ T_{xxyz} & -T_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xxyz} \\ 0 & 0 & 0 & 0 & T_{xxyz} & 0 \end{bmatrix}$$

$T_{xxyz} = \mathbb{T}_{4s}^{(1)}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & -T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -T_{xxyz} & T_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{xxyz} \\ 0 & 0 & 0 & 0 & T_{xxyz} & 0 \end{bmatrix}$$

$T_{xxyz} = 2\mathbb{M}_{3c}^{(1)}$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} T_{xxyz} & 0 & 0 \\ -T_{xxyz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -T_{xxyz} & 0 \end{bmatrix}$$

$T_{xxyz} = \mathbb{M}_{3c}^{(2)}$

* Rank 4 tensor (as).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$T_{yzxx} = \mathbb{M}_{3c}^{(3)}$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & T_{xxyz} & 0 & 0 & T_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & -T_{xxyz} & 0 & 0 & -T_{xxzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxx} & 0 & 0 & T_{yzxx} \\ 0 & 0 & 0 & 0 & T_{xxzy} & 0 & 0 & T_{xxyz} & 0 \end{bmatrix}$$

$$T_{xxyz} = 2\mathbb{M}_{3c}^{(1)} + \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)}$$
$$T_{xxzy} = 2\mathbb{M}_{3c}^{(1)} - \mathbb{M}_{3c}^{(2)} + \mathbb{T}_{4s}^{(1)}$$
$$T_{yzxx} = -2\mathbb{M}_{3c}^{(1)} + \mathbb{T}_{4s}^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} T_{yzxx} & -T_{yzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -T_{yzxx} & 0 & 0 & -T_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzxx} = \mathbb{M}_{3c}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{yyyz} \\ 0 & 0 & 0 \\ 0 & T_{yyyz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -T_{yyyz} \\ 0 & 0 & 0 \\ 0 & -T_{yyyz} & 0 \\ 0 & 0 & 0 \\ -T_{yyyz} & 0 & 0 \end{bmatrix}$$

$$T_{yyyz} = -\mathbb{T}_{4s}^{(1)}$$